

Customer Churn Analysis Dashboard

Project Overview

This project analyzes customer churn behavior in a telecommunications company using Python, SQL, and Power BI.

The objective was to identify factors influencing customer churn and provide actionable business recommendations to improve customer retention.

Tools Used

- Python (Pandas, NumPy, Matplotlib, Seaborn)
- SQL (MySQL)
- Power BI
- Excel

Key Analysis Performed

- Data Cleaning and Preparation
- Exploratory Data Analysis (EDA)
- Customer Segmentation
- SQL-Based Business Analysis
- Interactive Dashboard Development

Key Insights

- Month-to-month contract customers exhibited the highest churn rates.
- Customers without Tech Support and Online Security services were more likely to leave.
- Lower-tenure customers showed significantly higher churn tendencies.
- Fiber Optic customers represented a major churn-risk segment.

Business Recommendations

- Encourage long-term contracts through incentives.
- Promote Tech Support and Online Security services.
- Improve onboarding for new customers.
- Investigate churn drivers among Fiber Optic users.

Dashboard Pages

1. Executive Summary
2. Customer Demographics
3. Service Analysis
4. Revenue & Contract Analysis
5. Churn Risk & Recommendations